

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12405127
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Kawatani; Takeshi

Apparatus information display device

Abstract

An information display device of an apparatus includes: a standing member that stands upward in front of an operation position and has a lower portion which is supported by an apparatus main body; and a light guide member that extends from an apparatus main body side toward an upper portion side of the standing member, is arranged along a predetermined portion of the standing member, and causes at least part of the upper portion side of the standing member to emit light.

Inventors:	Kawatani; Takeshi (Tokyo, JP)
Applicant:	HONDA MOTOR CO., LTD. (Tokyo, JP)
Family ID:	88193836
Assignee:	HONDA MOTOR CO., LTD. (Tokyo, JP)
Appl. No.:	18/127697
Filed:	March 29, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230314155 A1	Oct. 05, 2023

Foreign Application Priority Data

JP	2022-061095	Mar. 31, 2022
----	-------------	---------------

Publication Classification

Int. Cl.: G01C21/36 (20060101); B60K35/22 (20240101); B60K35/28 (20240101); B60K35/60 (20240101); B60K35/65 (20240101); B60K35/81 (20240101); B60K35/215 (20240101)

U.S. Cl.:

CPC **G01C21/3632** (20130101); **B60K35/22** (20240101); **B60K35/28** (20240101);
B60K35/60 (20240101); **B60K35/654** (20240101); **B60K35/81** (20240101);
B60K35/215 (20240101)

Field of Classification Search

CPC: B60K (35/22); B60K (35/28); B60K (35/60); B60K (35/654); B60K (35/81); B60K
(35/215); G01C (21/3632)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10543812	12/2019	Green	N/A	H04N 23/57
2012/0274458	12/2011	Wagner	N/A	N/A
2017/0329143	12/2016	Svarichevsky	N/A	G06F 3/011

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
206561808	12/2016	CN	N/A
2002-350144	12/2001	JP	N/A
2010-236894	12/2009	JP	N/A
2020-001454	12/2019	JP	N/A
2020001454	12/2019	JP	N/A

OTHER PUBLICATIONS

Japanese Office Action for Japanese Patent Application No. 2022-061095 mailed Jul. 25, 2023.
cited by applicant

Primary Examiner: Han; Charles J

Attorney, Agent or Firm: Amin, Turocy & Watson, LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) Priority is claimed on Japanese Patent Application No. 2022-061095, filed on Mar. 31, 2022, the contents of which are incorporated herein by reference.

BACKGROUND

Field of the Invention

(2) The present invention relates to an information display device of an apparatus.

Background

(3) In the related art, a configuration is known which displays navigation information on a meter device of a vehicle (for example, refer to Japanese Unexamined Patent Application, First Publication No. 2002-350144).

SUMMARY

(4) However, in the above-described configuration of the related art, a driver alternately views a

vehicle forward direction (travel direction) and the meter device, and there is a problem in that the movement of the eye line is frequent and leads to fatigue.

(5) An aspect of the present invention provides an information display device of an apparatus capable of preventing the movement of an eye line and displaying information to an operator of the apparatus.

(6) An apparatus information display device according to a first aspect of the present invention is an information display device of an apparatus that is operated by an operator, including: a standing member that stands upward in front of an operation position of the apparatus and has a lower portion which is supported by an apparatus main body; and a light guide member that extends from an apparatus main body side toward an upper portion side of the standing member, is arranged along a predetermined portion of the standing member, and causes at least part of the upper portion side of the standing member to emit light.

(7) According to this configuration, by causing the upper portion side of the standing member located in the front of the operation position to emit light, it is possible to efficiently perform information display such as direction indication for the operator. Since the standing member stands at the front of the operation position of the apparatus, it is possible to perform information transmission to the operator while reducing the eye line movement for confirming information by the operator.

(8) In a second aspect, the standing member may be a transparent or translucent panel member, and the light guide member may be constituted of a transparent or translucent optical fiber.

(9) According to this configuration, even when the light guide member is arranged on a transparent standing member such as a windscreen of a vehicle, the light guide member can be inconspicuous, and it is possible to maintain a good appearance.

(10) In a third aspect, the light guide member may be arranged to be inserted in the inside in a horizontal cross-section of the standing member.

(11) According to this configuration, the light guide member can be inconspicuous, and it is possible to improve the appearance. Both the front and rear surfaces of the standing member when seen from the operation position can be easily maintained to be smooth. It is possible to prevent disturbance from reaching the light guide member.

(12) In a fourth aspect, the light guide member may be arranged to be exposed to any one of a first surface of the standing member that faces an operation position side and a second surface of the standing member that faces an opposite side of the operation position.

(13) According to this configuration, the light guide member can be easily arranged, and it is possible to facilitate manufacturing of the standing member that incorporates the light guide member.

(14) In a fifth aspect, the light guide member may be arranged along the outer circumferential edge of the standing member.

(15) According to this configuration, the light guide member can be easily arranged, and it is possible to facilitate manufacturing of the standing member that incorporates the light guide member. Both the front and rear surfaces of the standing member when seen from the operation position can be maintained to be smooth.

(16) In a sixth aspect, a light source that causes a light emission portion of the standing member to emit light via the light guide member may be arranged on the apparatus main body.

(17) According to this configuration, it is possible to reduce the weight of the standing member and reduce shaking, vibration, and the like of the standing member compared to a case where the light source is arranged on the standing member. Even in a configuration where the standing member moves upward and downward, the load of an actuator can be reduced.

(18) In a seventh aspect, the apparatus may be a saddle riding vehicle on which a driver rides by straddling a vehicle body, and the standing member may constitute a windscreen that is arranged on an upper portion of a front cowl at a vehicle body front part.

(19) According to this configuration, the information display device can be used as an indicator lamp such as a winker and a driving assist system included in the saddle riding vehicle, or can be used as a display device of a travel direction of a navigation system. As a result, information provided to the driver during traveling of the vehicle can be provided to the driver with a little eye line movement.

(20) In an eighth aspect, the apparatus may be an automobile on which a driver rides, the automobile may have an output device that outputs navigation information, and a plurality of light emission portions of the standing member may be provided to be distributed to right and left sides of a vehicle body, be arranged symmetrically in a rightward-leftward direction with respect to a vehicle body rightward-leftward middle, and emit light in response to the navigation information.

(21) According to this configuration, information (for example, right-left turn information of a course) relating to the right and left sides of the vehicle body in the navigation information can be easily confirmed with a little eye movement.

(22) In a ninth aspect, the navigation information may be right-left turn information in a travel direction of the automobile, and the light emission portion may display the right-left turn information in response to the navigation information, and vehicle speed and position information of the automobile.

(23) According to this configuration, for example, when the vehicle speed of the automobile is fast, the right-left turn information can be displayed at an early timing from a position in front of and sufficiently away from a right-left turn position, and when the vehicle speed of the automobile is slow, the right-left turn information can be displayed at a late timing from a position relatively close to a right-left turn position. In this way, the light emission timing of the right-left turn information can be optimized in response to the vehicle speed and the position information of the automobile.

(24) According to an aspect of the present invention, it is possible to provide an information display device of an apparatus capable of preventing the movement of an eye line and displaying information to an operator of the apparatus.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a left side view of a motorcycle in an embodiment of the present invention.

(2) FIG. 2 is a view showing an information display device that is provided on a windscreen of the motorcycle described above.

(3) FIG. 3 is a block diagram of a vehicle system that includes the information display device.

(4) FIG. 4 is a IV-IV cross-sectional view of FIG. 2.

(5) FIG. 5 is a cross-sectional view corresponding to FIG. 4 and showing a modification example of the embodiment.

(6) FIG. 6 is a cross-sectional view corresponding to FIG. 4 and showing a second modification example of the embodiment.

(7) FIG. 7 is a view corresponding to FIG. 2 and showing a third modification example of the embodiment.

DESCRIPTION OF EMBODIMENTS

(8) Hereinafter, an embodiment of the present invention will be described with reference to the drawings. Directions such as forward, rearward, rightward, and leftward directions in the following description are the same as those in a vehicle described below unless otherwise stated. In appropriate positions in the drawings used in the following description, an arrow FR that indicates a vehicle forward direction, an arrow LH that indicates a vehicle leftward direction, an arrow UP that indicates a vehicle upward direction, and a line CL that indicates a vehicle body rightward-

leftward middle are shown.

(9) <Entire Vehicle>

(10) FIG. 1 shows a motorcycle (apparatus) **1** of the present embodiment. The motorcycle **1** includes a front wheel **3** that is steered by a handle (operation handle) **2** and a rear wheel **4** that is driven by a power unit **U**. The motorcycle **1** is a saddle riding vehicle on which a driver (operator) rides by straddling a vehicle body and is able to swing (bank) the vehicle body in a rightward-leftward direction (roll direction) with reference to ground contact points of the front and rear wheels **3**, **4**. The vehicle of the embodiment includes not only a vehicle that rotates in a direction in which the vehicle body is banked like the motorcycle **1** but also a vehicle that rotates by the steering of a steering wheel without banking the vehicle body.

(11) The front wheel **3** of the motorcycle **1** is supported by a front wheel suspension device **3A**. The front wheel suspension device **3A** is supported by a front end block **6** of a vehicle body frame **5**. A bar handle **2** for steering the front wheel rotation is arranged above the front wheel suspension device **3A**. A pair of right and left main frames **7** extend diagonally downward and rearward at a rearward position of the front end block **6** of the vehicle body frame **5**. Each of rear end portions of the right and left main frames **7** is connected to each of a pair of right and left pivot frames **8**. A power unit **U** that includes, for example, an engine (internal combustion engine) is provided at a lower position of the right and left main frames **7** and a forward position of the right and left pivot frames **8**.

(12) A front end portion of a swing arm (not shown) is supported by the right and left pivot frames **8**. The rear wheel **4** of the motorcycle **1** is supported by a rear end portion of the swing arm. A rear cushion (not shown) is provided between the swing arm and the vehicle body frame **5**.

(13) A rear frame **9** is connected to rear portions of the right and left main frames **7** and the right and left pivot frames **8**. A seat **11** for seating an occupant is arranged above the rear frame **9**. A rear storage box **12** is arranged at a rearward position of the seat **11**. Right and left saddle bags **13** are arranged on right and left sides below the rear storage box **12**.

(14) A vehicle body front part **14A** of the motorcycle **1** is covered by a front cowl **14**. A windscreen **15** is arranged above a front portion of the front cowl **14**. The windscreen includes a screen main body **16** (a standing member, a panel member) made of a transparent or translucent resinous material such as a polycarbonate resin. The screen main body **16** stands upward from a vehicle body side (lower end portion side) at a forward position of a driving position (operation position) **11a** at which the driver performs driving of the motorcycle **1**. The screen main body **16** extends diagonally rearward and upward from the vehicle body side.

(15) An upper end height of the screen main body **16** is equivalent to an eye height of the driver. The driver can view a vehicle forward direction through the screen main body **16**.

(16) The screen main body **16** is constituted, for example, to be capable of moving upward and downward by a drive force of an electric actuator. A lower portion of the screen main body **16** is supported by the vehicle body front part (apparatus main body) **14A** that includes the front cowl **14** and the vehicle body frame **5** via an arm or the like which extends from the electric actuator.

(17) A meter device **17** is arranged at a forward position of the handle **2** and inside the front cowl **14**. The meter device **17** is, for example, separated from a steering-system component and is supported by the front cowl **14** or the vehicle body frame **5**. The meter device **17** includes, for example, a meter of a vehicle speed, an engine rotation speed, and the like, and a display screen **18** (refer to FIG. 3) such as a liquid crystal display that displays an image of various information. A meter visor **19** that extends diagonally rearward and upward along the windscreen **15** is arranged on an upper portion of the meter device **17**.

(18) <Information Display Device>

(19) Next, an information display device **21** that includes a light emission portion **22** and is arranged on the screen main body **16** is described.

(20) As shown in FIG. 1 to FIG. 4, the information display device **21** includes: a pair of right and

left light emission portions **22** that are arranged on an upper end edge (a standing front end edge, an end surface) **16c** of the screen main body **16**; a light unit **24** that is arranged on the vehicle body front part **14A** and includes a light source **23** and the like; and a light guide member **25** that extends upward from the light unit **24** to arrive at the right and left light emission portions **22** at an upper end of the screen main body **16** and guides light of the light source **23** to the light emission portion **22**.

(21) The left light emission portion **22** is arranged on a left side of the vehicle body rightward-leftward middle CL. The right light emission portion **22** is arranged on a right side of the vehicle body rightward-leftward middle CL. The right and left light emission portions **22** are arranged symmetrically in a rightward-leftward direction with respect to the vehicle body rightward-leftward middle CL. The right and left light emission portions **22** are controlled to perform a predetermined light emission display (lighting or blinking). The light emission of the right and left light emission portions **22** notifies the driver of predetermined information. The predetermined information includes, for example, winker operation information by light emission that interlocks with right and left wipers, right-left turn information of a travel direction in response to the navigation information or the like, notification and warning information relating to a driving assist control, an automated driving control, and the like of the motorcycle **1**, and the like.

(22) The light unit **24** is supported by the front cowl **14** or the vehicle body frame **5** at the inside of the front cowl **14**. The light unit **24** includes the light source **23** of the right and left light emission portions **22** and an electric power supply circuitry (driver). The light sources **23** of the right and left light emission portions **22** are provided separately from each other. Hereinafter, the light sources **23** of the right and left light emission portions **22** are referred to as right and left light sources **23**. Each of the right and left light sources **23** is, for example, an LED but may be an incandescent bulb. The emission light of each of the right and left light sources **23** is separately guided to each of the right and left light emission portions **22** by the light guide member **25** using an optical fiber.

(23) By arranging the light unit **24** that includes the light source **23** on the vehicle body side (for example, a back side of the meter device **17** when seen from the driving position **11a**), the weight of the screen main body **16** is reduced compared to the case where the light source **23** is arranged on the screen main body **16**. In an electric windscreen **15**, the load of the electric actuator can be reduced. Compared to the case where the light source **23** is arranged on the screen main body **16**, the light source **23** does not enter the field of view of the driver, and it is possible to prevent the driver from becoming distracted. The light unit **24** does not interfere with the display of the meter device **17**.

(24) The light guide member **25** defines a form, for example, as an optical fiber cable formed of a plurality of bundled optical fiber core wires. The optical fiber is made of a material having a high refractive index such as transparent glass or a transparent resinous material such as an acrylic resin or a polycarbonate resin. The optical fiber and the light guide member **25** have flexibility. The light guide member **25** is transparent or translucent and is less conspicuous even when arranged on a transparent or translucent screen main body **16**. The light guide member **25** may be a molded component that is formed to be along a predetermined portion of the screen main body **16**.

(25) In the motorcycle **1**, an occupant (driver) seated on the seat **11** drives the motorcycle **1** while looking in the vehicle forward direction. The windscreen **15** is arranged at a forward position the driving position **11a** above the seat **11**. The windscreen **15** is located at a height which enters the field of view of the driver when the driver seated on the seat **11** sees the vehicle forward direction. The light guide member that is arranged on the windscreen **15** (screen main body **16**) becomes inconspicuous, and thereby, it is possible to prevent the driver from becoming distracted.

(26) The light guide member **25** extends from the light unit **24** that is arranged on the vehicle body front part **14A** side. A pair of right and left light guide members **25** are provided to extend from the right and left light sources **23** to the right and left light emission portions **22**. The screen main body **16** defines a gently curved shape that protrudes forward in plan view. The right and left light guide

members **25** extend in an upward-downward direction so as to form a virtual bend line on right and left side portions of the screen main body **16**. Each of the right and left light guide members **25** is arranged in a straight line form to an upper portion from a lower portion of the screen main body **16**. The right and left light guide members **25** are inserted into an insertion portion **26** formed in a horizontal cross-section of the screen main body **16**.

(27) Reference numeral **16b** in the drawings indicates a surface (rear surface) on the front side of the screen main body **16** when seen from the driving position **11a**, and reference numeral **16a** in the drawings indicates a surface (front surface) on the back side of the screen main body **16** when seen from the driving position **11a**. In the horizontal cross-section of the screen main body **16**, for example, a front surface projection portion **26a** that projects forward from the front surface **16a** (second surface) is formed on the front surface **16a** side of the insertion portion **26**, and a rear surface projection portion **26b** that projects rearward from the rear surface **16b** (first surface) is formed on the rear surface **16b** side of the insertion portion **26**. The light guide member **25** is arranged along the front and rear surfaces **16a**, **16b** of the screen main body **16**.

(28) The light guide member **25** has an incident surface at one end in a longitudinal direction. The incident surface faces and is adjacent to the light source **23** (LED) from an irradiation direction. Light of the light source **23** enters the incident surface, and the light guide member **25** guides the light along the longitudinal direction. In the light guide member **25**, most of the light introduced from the incident surface is totally reflected within the light guide member **25**. The incident light into the light guide member **25** is guided along the longitudinal direction of the light guide member **25** while repeating reflection within the light guide member **25**. Another end in the longitudinal direction of the light guide member **25** is a light emission surface, and the light introduced into the light guide member **25** is emitted from the light emission surface. The light emission surface of the light guide member **25** reaches an upper end edge **16c** of the screen main body **16**. A light emission portion **22** that diffuses the light emission of the light emission surface of the light guide member **25** to a predetermined shape is arranged on both right and left sides of the upper end edge **16c** of the screen main body **16**.

(29) The light emission portion **22** is not limited to a configuration that is arranged on the upper end edge **16c** of the screen main body **16** and may be arranged on an upper end portion that includes a width of about 10 to 20 mm from the upper end edge **16c**. The light emission portion **22** may be arranged in an upper range (upper portion) than the middle in the upward-downward direction of the screen main body **16**.

(30) When the upper end edge **16c** is curved, for example, a portion having an angle of less than 45 degrees with respect to the horizontal plane is defined as the upper end edge **16c**. The upper end edge **16c** is provided in a range having a predetermined width in the rightward-leftward direction.

(31) By arranging the light guide member **25** to be inserted into the cross-section of the screen main body **16**, the light guide member **25** becomes inconspicuous, the front and rear surfaces **16a**, **16b** of the screen main body **16** can be easily maintained to be smooth, and it is possible to prevent disturbance from reaching the light guide member **25**.

(32) The example of FIG. 5 shows a form in which a groove **27b** for arranging the light guide member **25** is formed on the rear surface **16b** of the screen main body **16**, and the light guide member **25** is fitted into the groove **27b** and fixed.

(33) Further, the example of FIG. 6 shows a form in which a groove **27a** for arranging the light guide member **25** is formed on the front surface **16a** of the screen main body **16**, and the light guide member **25** is fitted into the groove **27a** and fixed.

(34) The grooves **27a**, **27b** in which the light guide member **25** is fitted may be sealed by an adhesive or the like. However, regardless of the presence or absence of the sealing, the forms of FIG. 5 and FIG. 6 show that the light guide member **25** is arranged to be exposed to the front surface **16a** side or the rear surface **16b** side of the screen main body **16**.

(35) In the forms of FIG. 5 and FIG. 6, it is conceivable that the smoothness on the front surface

16a side or the rear surface **16b** side of the screen main body **16** may be affected, and the appearance or the like may be affected; however, a long light guide member **25** can be easily arranged along the front and rear surfaces **16a**, **16b** of the screen main body **16**, and it is possible to facilitate manufacturing of the screen main body **16** that incorporates the light guide member **25**.
(36) The example of FIG. 7 shows a form in which the light guide member **25** is arranged along the outer circumferential edge **28** of the screen main body **16** when seen from a vehicle forward-rearward direction. The outer circumferential edge **28** includes an upper end edge **16c** and right and left side edges **28a**.

(37) In the form of FIG. 7, a long light guide member **25** can be further easily arranged, and it is possible to further facilitate manufacturing of the screen main body **16** that incorporates the light guide member **25**. The grooves **27a**, **27b** and the light guide member **25** shown in FIG. 5 and FIG. 6 are eliminated from the front and rear surfaces **16a**, **16b** of the screen main body **16**, and it is possible to maintain the front and rear surfaces **16a**, **16b** to be smooth.

(38) <System Configuration>

(39) FIG. 3 is a block diagram of a vehicle system that includes the information display device **21** in the embodiment.

(40) The motorcycle **1** includes a meter device **17**, an information display device **21** (light unit **24**), a navigation device **31**, a communication device **32**, an ETC (electronic toll collection system) in-vehicle device **33**, and a control device **34**.

(41) The navigation device **31** communicates with the outside and thereby performs route guidance or the like for the occupant.

(42) The communication device **32** is a portable terminal that is owned by the occupant and is capable of communicating with the outside and is a so-called smartphone, a tablet terminal, or the like. The communication device **32** includes an application (a software module that provides various functions to a user, that is, an occupant of the motorcycle **1**) having various functions such as a navigation function.

(43) The ETC in-vehicle device **33** exchanges toll information with a tollbooth of a toll road when the motorcycle **1** travels on the toll road.

(44) Information from the navigation device **31**, the communication device **32**, and the ETC in-vehicle device **33** are input to the control device **34**, and the control device **34** controls display of each of the meter device **17** and the information display device **21**. The control device **34** is an ECU (electronic control device) and realizes various functions by executing a program stored in a memory (not shown). Various sensors **35** and various switches **36** are connected to the control device **34**.

(45) <Information Display Example>

(46) The information display device **21** of the embodiment displays guidance information (right-left turn information of the travel direction) in conjunction with the navigation device **31**. For example, when the travel direction of the motorcycle **1** is a left turn direction at a predetermined guidance point, the control device **34** lights the light source **23** on the left side and lights or blinks the light emission portion **22** on the left side at the upper end of the screen main body **16** on the basis of route guidance information that is input from the navigation device **31**. When the travel direction of the motorcycle **1** is a right turn direction at a predetermined guidance point, the control device **34** lights the light source **23** on the right side and lights or blinks the light emission portion **22** on the right side at the upper end of the screen main body **16**. When the motorcycle **1** turns right or left and the guidance contents are executed, the control device **34** stops the lighting of the light source **23** and switches the light emission portion **22** to be in an OFF state.

(47) For example, when the motorcycle **1** does not execute the right-left turn guidance contents (when passing), light emission such as simultaneous blinking of the right and left light emission portions **22** may be performed, and the driver may be alerted.

(48) Further, when the motorcycle **1** is approaching a guidance point of a right-left turn, light

emission such as simultaneous blinking of the right and left light emission portions **22** may be performed, and the driver may be notified of approaching the guidance point.

(49) Alternatively, a plurality of light emission portions **22** may be provided on each of right and left sides, and the number of light emissions may be increased or decreased depending on the distance to the guidance point.

(50) Information (navigation information) from the navigation device **31** is output, for example, to the display screen **18** (output device) of the meter device **17**. The light emission of the right and left light emission portions **22** of the information display device **21** is performed in accordance with the information (for example, right-left turn information of the course) relating to the vehicle body right and left sides in the navigation information.

(51) The light emission of the right and left light emission portions **22** may display the right-left turn information of the course by additionally using the vehicle speed and the position information of the motorcycle **1** in addition to the navigation information.

(52) That is, the information display device **21** may display the right-left turn information at an early timing from a position in front of and sufficiently away from the right-left turn position when the vehicle speed of the motorcycle **1** is fast, and may display the right-left turn information at a late timing from a position relatively close to the right-left turn position when the vehicle speed of the motorcycle **1** is slow. In this way, by changing the light emission timing of the right-left turn information in accordance with the vehicle speed and the position information of the motorcycle **1**, it is possible to provide the right-left turn information with a margin and also optimize the light emission time of the right and left light emission portions **22**.

(53) There may be a configuration in which the motorcycle **1** does not include the navigation device **31**. That is, a configuration may be employed in which an application of the communication device **32** performs the route guidance. A configuration may be employed in which the motorcycle **1** is connectable to (capable of communicating with) the communication device **32** in a wired or wireless manner, and the navigation information received from the communication device **32** is output to the meter device **17** and is displayed on the display screen **18**.

(54) As described above, the information display device **21** of the apparatus in the embodiment described above includes: the screen main body **16** that stands upward in front (at a forward position) of the driving position **11a** of the motorcycle **1** and has a lower portion which is supported by the vehicle body front part **14A**; and the light guide member **25** that extends from the vehicle body front part **14A** side toward the upper portion side of the screen main body **16**, is arranged along the predetermined portion of the screen main body **16**, and causes at least part of the upper portion side of the screen main body **16** to emit light.

(55) According to this configuration, by causing the upper portion side of the screen main body **16** located in the front of the driving position **11a** to emit light, it is possible to efficiently perform information display such as direction indication for the driver. Since the screen main body **16** stands at the front of the driving position **11a** of the motorcycle **1**, it is possible to perform information transmission to the driver while reducing the eye line movement for confirming information by the driver.

(56) Further, in the information display device **21** of the apparatus described above, the screen main body **16** is a transparent or translucent panel member, and the light guide member **25** is constituted of a transparent or translucent optical fiber.

(57) According to this configuration, even when the light guide member **25** is arranged on the transparent screen main body **16** of the motorcycle **1**, the light guide member **25** can be inconspicuous, and it is possible to maintain a good appearance.

(58) Further, in the information display device **21** of the apparatus described above, the light guide member **25** is arranged to be inserted in the inside of the horizontal cross-section of the screen main body **16**.

(59) According to this configuration, the light guide member **25** can be inconspicuous, and it is

possible to improve the appearance.

(60) Both the front and rear surfaces of the screen main body **16** when seen from the driving position **11a** can be maintained to be smooth. It is possible to prevent disturbance from reaching the light guide member **25**.

(61) Further, in the information display device **21** of the apparatus described above, the light guide member **25** may be arranged to be exposed to any one of the rear surface **16b** of the screen main body **16** that faces the driving position **11a** side and the front surface **16a** of the screen main body **16** that faces the opposite side of the driving position **11a**.

(62) According to this configuration, the light guide member **25** can be easily arranged, and it is possible to facilitate manufacturing of the screen main body **16** that incorporates the light guide member **25**.

(63) Further, in the information display device **21** of the apparatus described above, the light guide member **25** may be arranged along the outer circumferential edge **28** of the screen main body **16**.

(64) According to this configuration, the light guide member **25** can be easily arranged, and it is possible to facilitate manufacturing of the screen main body **16** that incorporates the light guide member **25**. Both the front and rear surfaces of the screen main body **16** when seen from the driving position **11a** can be maintained to be smooth.

(65) Further, in the information display device **21** of the apparatus described above, the light source **23** that causes the light emission portion **22** of the screen main body **16** to emit light via the light guide member **25** is arranged on the vehicle body front part **14A**.

(66) According to this configuration, it is possible to reduce the weight of the screen main body **16** and reduce shaking, vibration, and the like of the screen main body **16** compared to a case where the light source **23** is arranged on the screen main body **16**. Even in a configuration where the screen main body **16** moves upward and downward, the load of an actuator can be reduced.

(67) Further, in the information display device **21** of the apparatus described above, the screen main body **16** constitutes the windscreen **15** that is arranged on the upper portion of the front cowl **14** at the vehicle body front part **14A** of the motorcycle **1**.

(68) According to this configuration, the information display device **21** can be used as an indicator lamp such as a winker and a driving assist system included in the motorcycle **1**, or can be used as a display device of a travel direction of a navigation system. As a result, information provided to the driver during traveling of the vehicle can be provided to the driver with a little eye line movement.

(69) Further, in the information display device **21** of the apparatus described above, the motorcycle **1** has an output device (the display screen **18**) that outputs navigation information, and a plurality of light emission portions **22** of the screen main body **16** are provided to be distributed to right and left sides of the vehicle body, are arranged symmetrically in the rightward-leftward direction with respect to the vehicle body rightward-leftward middle CL, and emit light in response to the navigation information.

(70) According to this configuration, information (for example, right-left turn information of a course) relating to the right and left sides of the vehicle body in the navigation information can be easily confirmed with a little eye movement.

(71) Further, in the information display device **21** of the apparatus described above, the navigation information is right-left turn information in the travel direction of the motorcycle **1**, and the light emission portion **22** displays the right-left turn information in response to the navigation information, and the vehicle speed and the position information of the motorcycle **1**.

(72) According to this configuration, for example, when the vehicle speed of the motorcycle **1** is fast, the right-left turn information can be displayed at an early timing from a position in front of and sufficiently away from a right-left turn position, and when the vehicle speed of the motorcycle **1** is slow, the right-left turn information can be displayed at a late timing from a position relatively close to a right-left turn position. In this way, the light emission timing of the right-left turn information can be optimized in response to the vehicle speed and the position information of the

motorcycle 1.

(73) The present invention is not limited to the embodiment described above. For example, the information display device of the apparatus of the embodiment is applied to motorcycles but may be applied to saddle riding vehicles other than motorcycles. Saddle riding vehicles include all vehicles on which a driver rides by straddling the vehicle body and include not only motorcycles (including motorized bicycles and scooter-type vehicles) but also three-wheeled vehicles (including vehicles having two front wheels and one rear wheel in addition to vehicles having one front wheel and two rear wheels) or four-wheeled vehicles (four-wheeled buggies or the like). The present invention may be applied to vehicles including an electric motor as a motor. The present invention may be applied to vehicles (passenger vehicles, buses, trucks, or the like) other than saddle riding vehicles.

(74) Further, the application of the present invention is not limited to vehicles. The present invention may be applied to various vehicles and movable bodies such as various transport apparatuses such as aircraft or ships, construction machinery, and industrial machinery. Further, the present invention is broadly applicable to apparatuses other than vehicles if the apparatus includes the standing member in the front of the operation position such as, for example, push mowers or cleaning machines.

(75) The configurations in the embodiment described above are merely examples of the present invention, and various changes can be made without departing from the scope of the present invention such as replacing the components of the embodiment with well-known components.

Claims

1. An apparatus information display device that is an information display device of an apparatus which is operated by an operator, comprising: a standing member that stands upward in front of an operation position of the apparatus and has a lower portion which is supported by an apparatus main body; and a light guide member that extends from an apparatus main body side toward an upper portion side of the standing member, is arranged along a predetermined portion of the standing member, and causes at least part of the upper portion side of the standing member to emit light, wherein a light emission portion that emits light using light guided by the light guide member is arranged on an upper end edge exposed to an operation position side in the standing member.
2. The apparatus information display device according to claim 1, wherein the standing member is a transparent or translucent panel member, and the light guide member is constituted of a transparent or translucent optical fiber.
3. The apparatus information display device according to claim 1, wherein the light guide member is inserted in an inside in a horizontal cross-section of the standing member.
4. The apparatus information display device according to claim 1, wherein the light guide member is exposed to any one of a first surface of the standing member that faces an operation position side and a second surface of the standing member that faces an opposite side of the operation position.
5. The apparatus information display device according to claim 1, wherein the light guide member is arranged along an outer circumferential edge of the standing member.
6. The apparatus information display device according to claim 1, wherein a light source that causes the light emission portion of the standing member to emit light via the light guide member is arranged on the apparatus main body.
7. The apparatus information display device according to claim 1, wherein the apparatus is a saddle riding vehicle on which a driver rides by straddling a vehicle body, and the standing member constitutes a windscreen that is arranged on an upper portion of a front cowl at a vehicle body front part.
8. The apparatus information display device according to claim 1, wherein the apparatus is an automobile on which a driver rides, the automobile has an output device that outputs navigation

information, and a plurality of light emission portions that are provided on the standing member are distributed to right and left sides of a vehicle body, are arranged symmetrically in a rightward-leftward direction with respect to a vehicle body rightward-leftward middle, and emit light in response to the navigation information.

9. The apparatus information display device according to claim 8, wherein the navigation information is right-left turn information in a travel direction of the automobile, and the light emission portion displays the right-left turn information in response to the navigation information, and vehicle speed and position information of the automobile.
